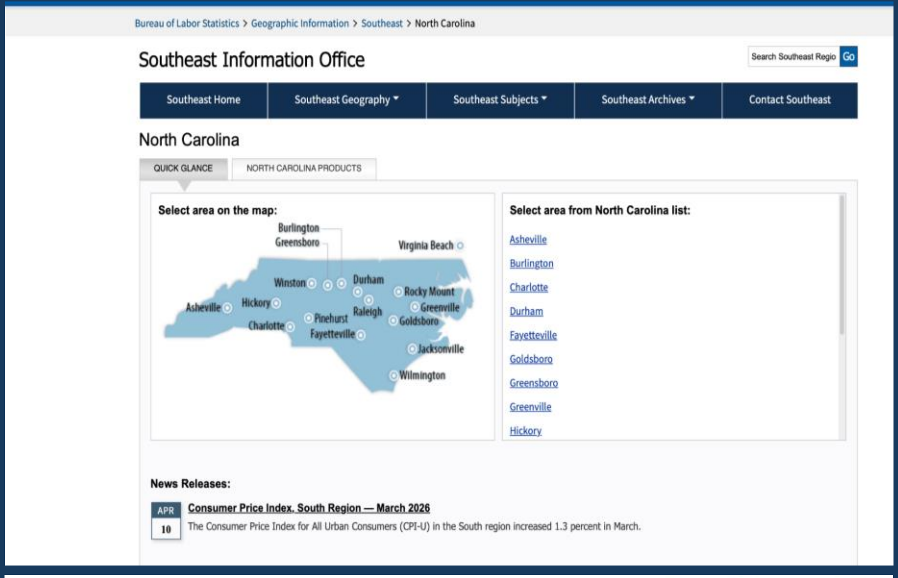
North Carolina Labor Market Trends

Analyzing unemployment, employment levels, wage growth, and economic shocks (2016-2026)

BLS labor market data

Tableau data analysis

Group: Claudia & the others
Course: BUS 446



Project Objective & Core Questions

What the project is about and why the analysis matters

Business objective
The goal of this project is to understand how key labor market variables interact and change over time. The analysis will focus on identifying patterns between employment, unemployment, and wage growth.

Q1 How has the unemployment rate in North Carolina changed from 2016 to 2026, and what overall trends can be observed in the labor market during this period ?

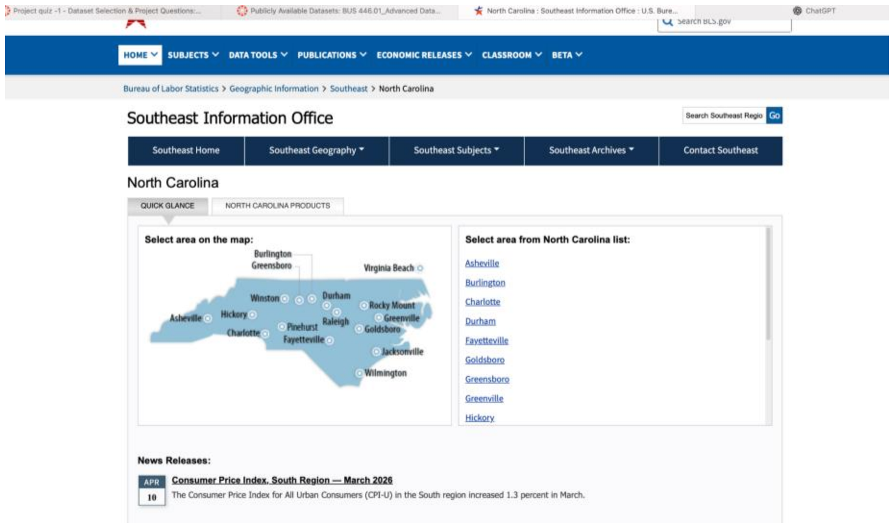
Q2 What relationship exists between employment levels and average hourly wages in North Carolina, and can increases in employment help explain wage growth ?

Q3 Are changes in unemployment associated with noticeable changes in employment levels and wages, particularly during periods of economic disruption or recovery ?

The value of the project is not only showing trends, but explaining what those trends imply for workforce planning, wage pressure, and economic resilience.

Dataset Introduction

Sources, variables, and limitations



Primary dataset

U.S. Bureau of Labor Statistics (BLS) data on North Carolina labor market trends.

Main variables

Year/month, unemployment rate, employment level, and average hourly wages.

Supporting data

Data.gov and cost-of-living context used to support and validate insights.

Limitations

Data is aggregated, may require cleaning, and may not reflect local-level differences.

Data sources and methods are documented in the GitHub README to ensure transparency and reproducibility.

Analytical Approach

From raw data to business conclusions

1 → 2 → 3 → 4 → 5

1. Collect data

BLS public datasets focused on North Carolina labor market

2. Clean & join

Standardized dates, units, and variables across datasets

3. Build Tableau views

Created trend lines and visuals for employment and wages

4. Regression test

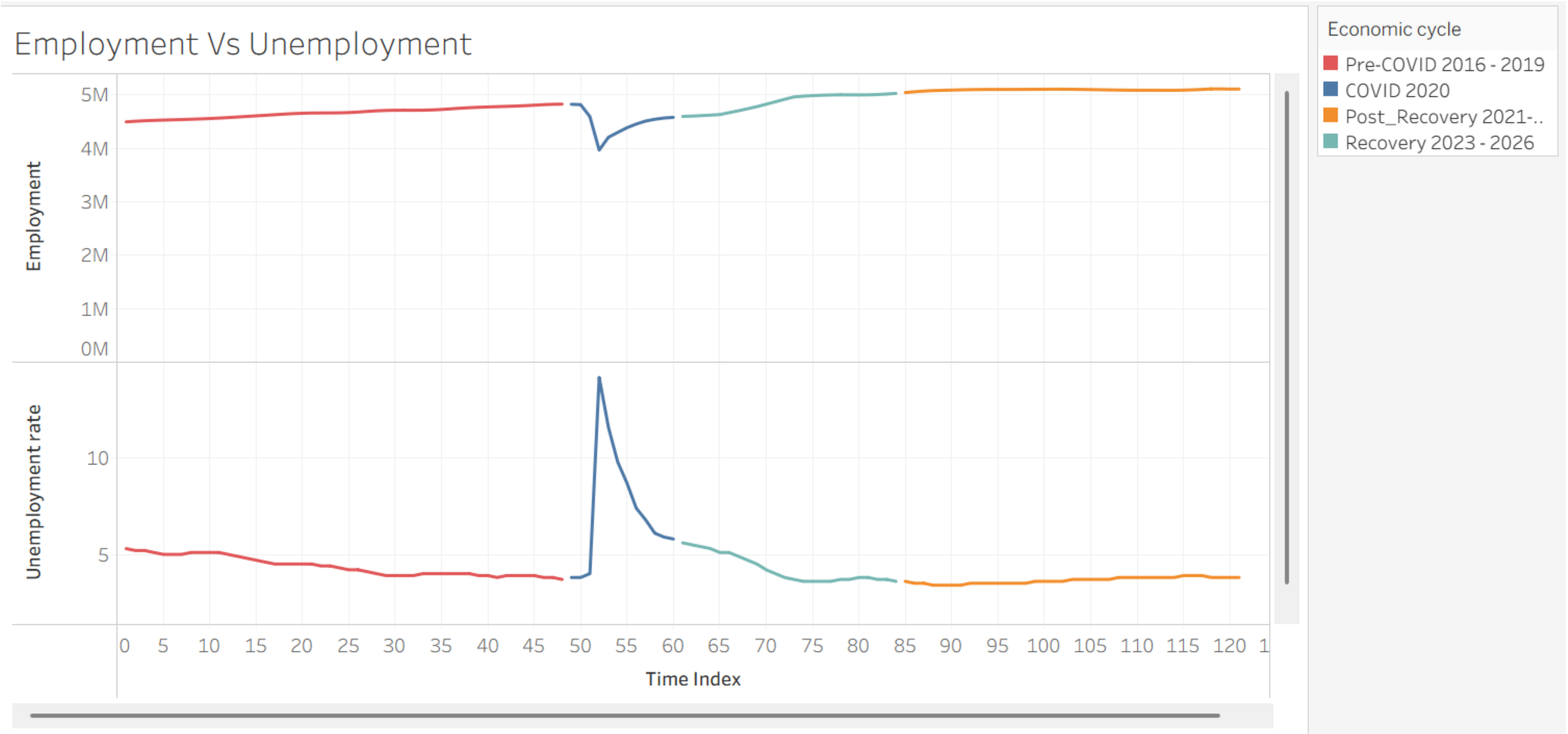
Measured how employment levels impact wage growth

5. Interpret insights

Connected results to real business implications

The analysis combines visualization and regression to explain how employment levels influence wage growth over time.

Tableau Dashboard 1: Labor Market Trends

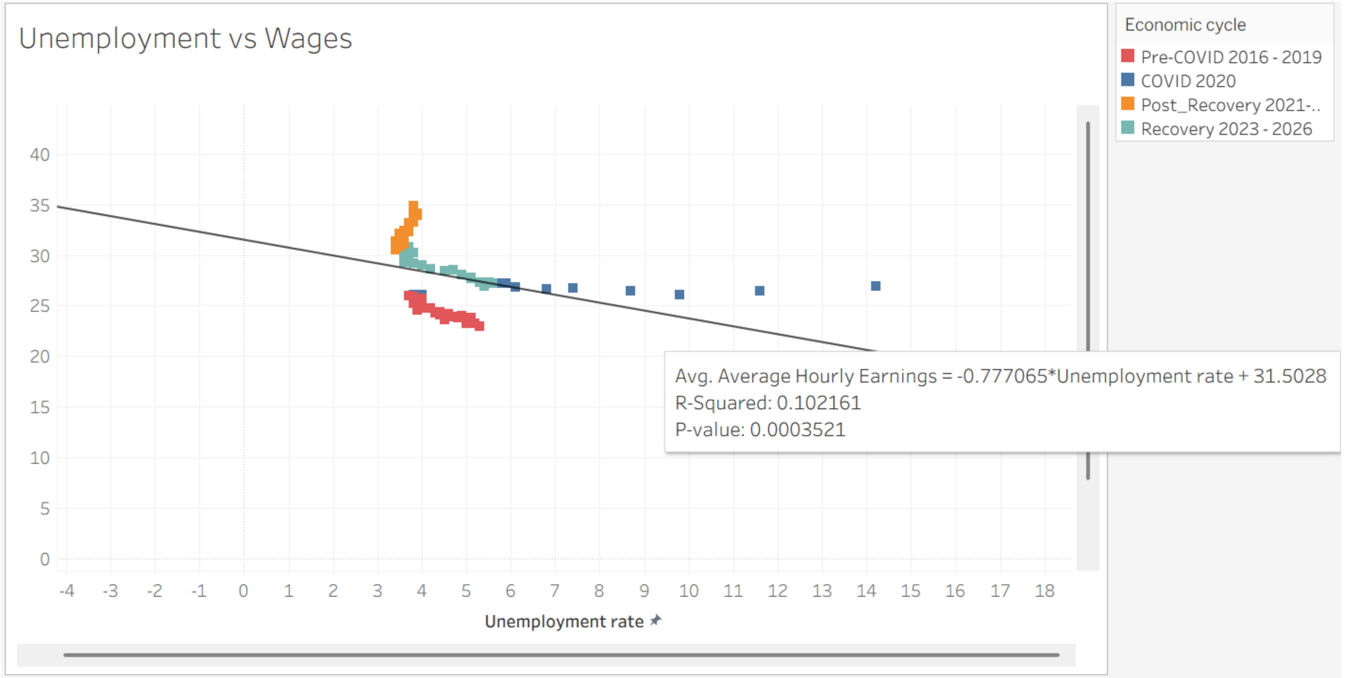


This chart tracks employment and unemployment in North Carolina from 2016-2026. The two variables move in opposite directions, consistent with standard labor market dynamics.

- **Pre-2020:** Stable employment with low unemployment
- **2020:** Sharp drop in employment and spike in unemployment (COVID-19 shock)
- **Post-2020:** Gradual employment recovery and declining unemployment

Research Question 1: Unemployment vs. Wages

How strongly are unemployment changes related to average hourly wages?



A negative relationship suggests that wages tend to be higher when labor markets are tighter (lower unemployment). In this data, wages increase as unemployment falls, but the relationship is relatively weak, indicating that wage growth is influenced by additional factors.

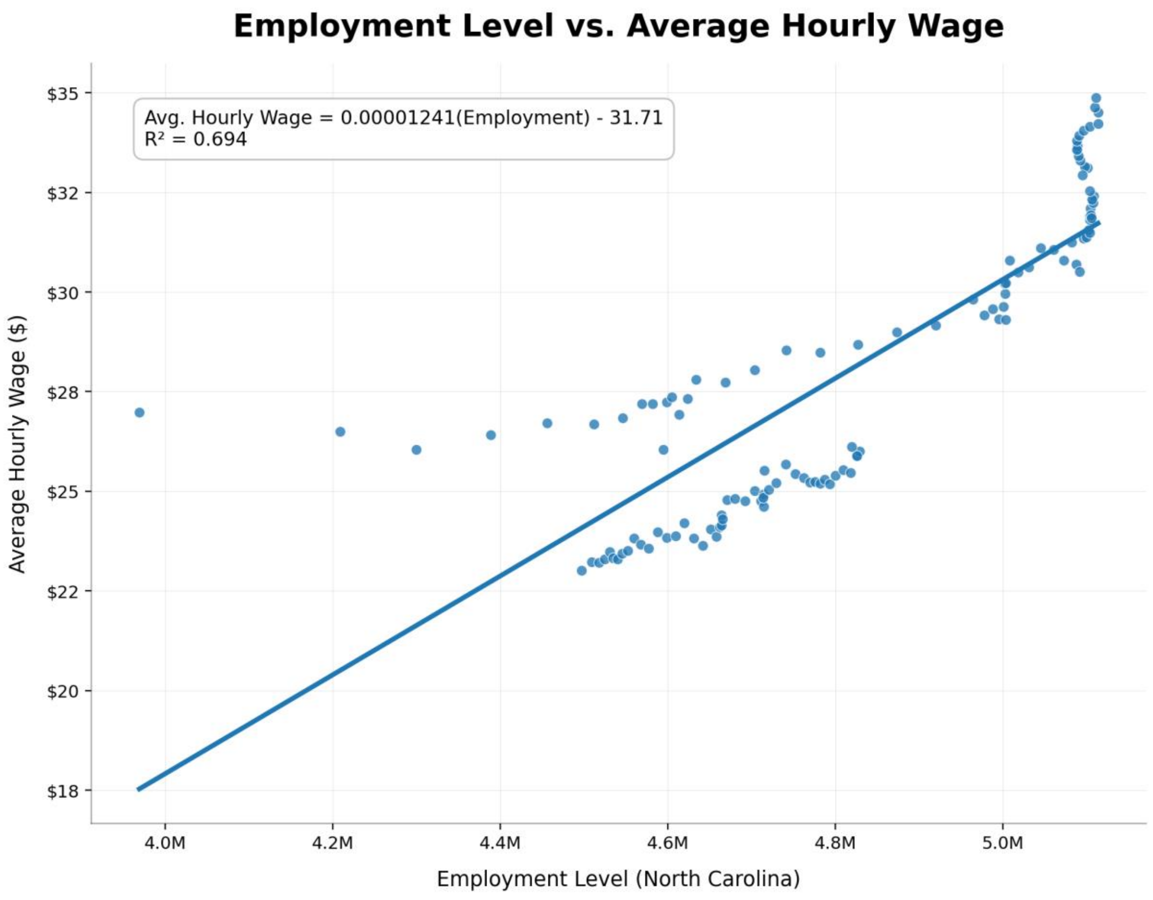
- **Trend-line direction:** Negative
- **Correlation:** Negative (implied by slope)
- **R^2:** 0.10
- **Statistical significance:** $p < 0.001$

The relationship is statistically significant but explains a limited share of wage variation.

From 2016–2026, unemployment fluctuated—rising sharply during 2020 and declining during recovery—while average hourly wages steadily increased. The relationship appears weak because unemployment explains only about 10% of wage variation, suggesting other economic forces play a larger role.

Research Question 2: Employment Levels & Wage Growth

Can employment help predict wage increases?



Regression model idea

$$\text{Avg. Hourly Wage} = \beta_0 + \beta_1(\text{Employment Level}) + \varepsilon$$

Interpretation

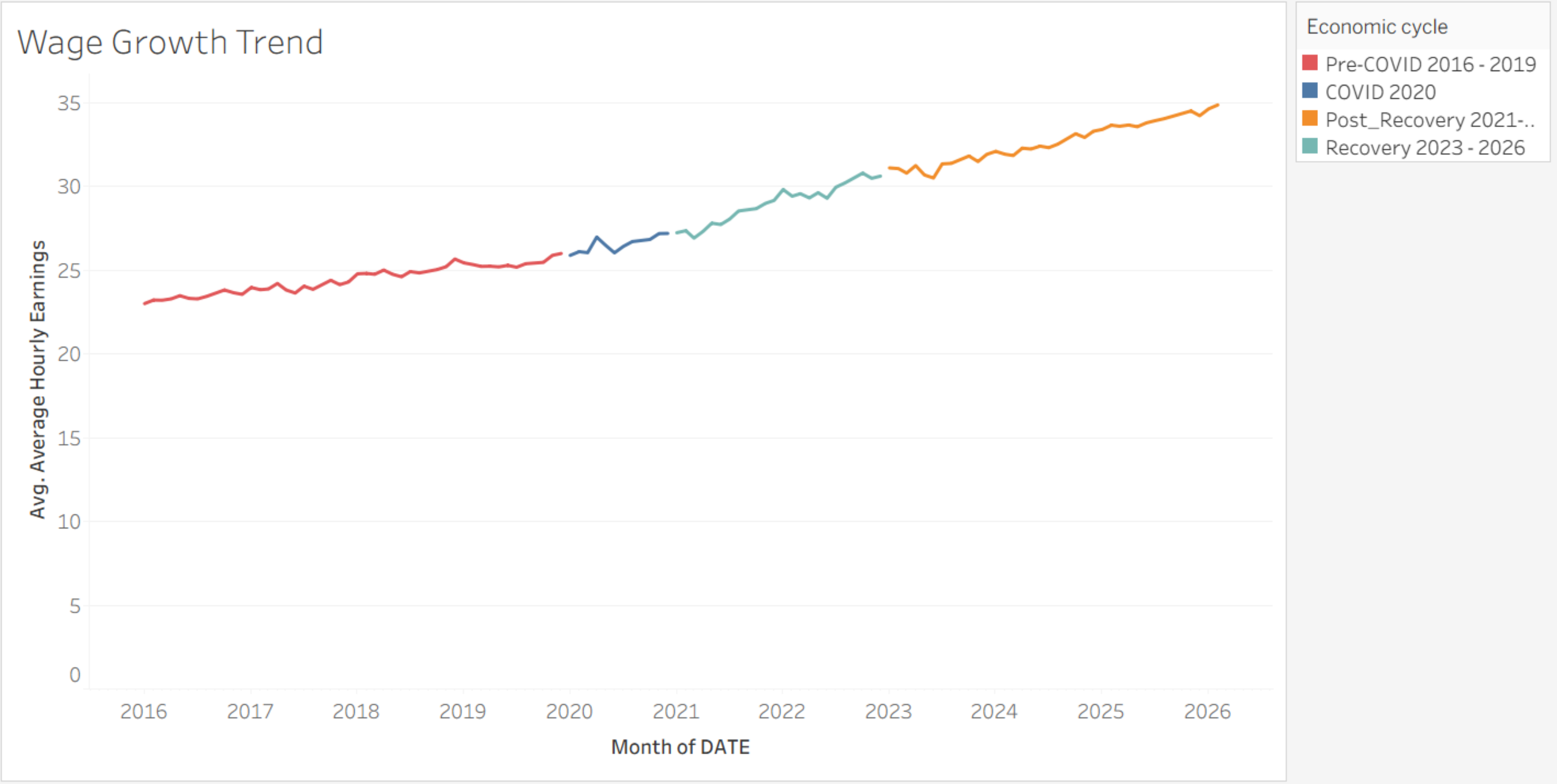
Positive relationship: higher employment → higher wages
Effect is meaningful but not perfect
Other factors also influence wage growth

Include from Excel

Slope (β_1): Positive
R^2: 0.694 (~69%)
Meaning: Employment strongly helps predict wages

Research Question 3: Wage Growth Trend

Do major shocks align with visible labor market shifts?



Annotation ideas
• 2020 COVID labor-market disruption
• Recovery period
• Wage acceleration or stabilization
• Employment rebound

Average hourly wages rose steadily across all periods, accelerating after 2020, suggesting wage growth continued despite COVID-related labor disruptions.

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Key Findings & Business Implications

Replace bracketed text with final numbers after Tableau/Excel analysis

Finding 1

- Unemployment decreased overall (2016-2026)
- Wages increased during the same period
- Moderate relationship between unemployment and wages

Finding 2

- Employment is positively related to wages
- Regression shows a strong relationship ($R^2 \approx$ 0.69)
- Employment meaningfully predicts wage growth

Finding 3

- Economic shocks caused noticeable shifts
- Biggest impact around 2020 (COVID period)
- Employment and wages were both affected

Implications for decision-makers

- Employers: expect wage increases when employment rises
- Workforce planners: track employment + unemployment together
- Analysts: use multiple indicators, not just one

Portfolio message

- Project is clear and easy to follow
- Data, visuals, and methods are well organized
- Conclusions are supported and easy to understand

Conclusion & Recommendations

What the North Carolina labor market analysis shows

Main conclusion

North Carolina’s labor market recovered from the 2020 shock and continued moving toward higher employment and higher wages.

The strongest wage signal in this project is employment growth—not unemployment alone.

Business recommendation

Employers and workforce planners should monitor employment, unemployment, and wage trends together when forecasting labor costs, hiring demand, and economic resilience.

1. **Employment matters most**

   Employment levels show a strong positive relationship with wages ($R^2 \approx 0.69$).

2. **Unemployment explains less**

   Unemployment is significant but only explains a limited share of wage variation ($R^2 \approx 0.10$).

3. **2020 was the key disruption**

   COVID created a sharp labor-market shock, followed by recovery and continued wage growth.

Final takeaway

A stronger labor market can create wage pressure, but wage growth is best understood through multiple indicators—not one variable alone.

Q&A

Open for questions